

Tarun Kumar Angalla

AI Engineer

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GitHub: <https://github.com/TarunAngalla>

Professional Summary

AI Engineer and AWS Certified Developer with 3+ years of experience building production-style LLM-powered systems, RAG architectures, AI agents, prompt frameworks, evaluation workflows, vector-search applications, and AI-enabled SaaS products. Experienced with Python, OpenAI APIs, Gemini API, Anthropic/Claude concepts, LangChain, LangGraph, Pinecone, FAISS, ChromaDB, PostgreSQL, FastAPI, AWS, Docker, GitHub Actions, MLflow, and structured output validation. Strong background designing system prompts, behavior definitions, retrieval grounding, response quality checks, regression tests, adversarial test cases, safety guardrails, and provider-comparison workflows. Skilled at translating non-technical product requirements such as tone, voice, boundaries, quality, and user experience into precise AI behavior specifications that engineering, product, creative, and brand teams can evaluate and improve.

Technical Skills

LLM Systems: OpenAI APIs, Gemini API, Anthropic/Claude Concepts, Multi-provider LLM Integration, Model Selection
Prompt Engineering: System Prompts, Few-shot Prompting, Multi-turn Conversation Design, Structured Outputs, Prompt Versioning
Character AI: Character Behavior Definitions, Personality Constraints, Voice/Tone Rules, Canon Grounding, Response Boundaries
RAG Architecture: Embeddings, Chunking, Vector Stores, Pinecone, FAISS, ChromaDB, Retrieval Evaluation, Relevance Scoring
Evaluation: LLM Output Evaluation, Golden Test Sets, Regression Testing, Adversarial Testing, Safety Testing, Quality Metrics
Guardrails: Guardrails AI Concepts, NeMo Guardrails Concepts, Policy Checks, Hallucination Reduction, Refusal Boundaries
Agentic AI: LangChain, LangGraph, Tool Calling, Workflow Orchestration, State Management, Failure Handling
Programming: Python, SQL, JavaScript, TypeScript, Bash, FastAPI, REST APIs, JSON, YAML
Cloud & DevOps: AWS, S3, SageMaker, EC2, Docker, GitHub Actions, CI/CD, Linux, Environment Configuration
Data & ML: TensorFlow, Keras, Scikit-learn, XGBoost, PyTorch Concepts, MLflow, NLP, Classification, RMSE, F1
Voice AI Concepts: TTS Concepts, STT Concepts, Spoken Response Design, Latency-aware Prompting, Conversational UX

Professional Experience

ZoomInfo Mar 2024 – Present
AI Engineer Remote, USA

- Built production-style LLM-powered workflows using Python, OpenAI APIs, LangChain, PostgreSQL, Databricks, Pinecone, REST APIs, Docker, GitHub Actions, and AWS deployment patterns.
- Designed system prompts, prompt templates, retrieval instructions, response schemas, and guardrail-style validation rules to control AI behavior across multi-step business workflows.
- Built RAG pipelines using embeddings, Pinecone vector indexing, metadata filtering, chunking strategies, retrieval logic, grounding checks, and relevance validation for fragmented enterprise knowledge.
- Developed prompt and response evaluation workflows using MLflow, golden examples, response review flags, grounding validation, failure-case tagging, regression checks, and quality review notes.
- Identified 22% fewer unsupported AI recommendations by improving system prompts, retrieval grounding, response schemas, validation logic, and human-review checkpoints.
- Improved retrieval performance by refining chunking, metadata filters, vector indexing, and query routing, reducing search latency by 27% while maintaining response relevance.
- Built reusable Python services for API calls, payload normalization, data transformation, retry handling, prompt input construction, structured output parsing, and response validation.
- Integrated AI workflows with backend services, SQL datasets, reporting layers, dashboards, and SaaS applications through REST APIs, JSON contracts, and documented API schemas.
- Designed agentic workflow patterns for research, summarization, evidence retrieval, recommendation generation, structured extraction, output validation, and exception handling.
- Created multi-provider-ready AI workflow patterns that separated provider configuration, prompt contracts, retrieval context, model parameters, output schemas, and validation rules.
- Supported cost-aware and latency-aware LLM workflows by reducing redundant context retrieval, shortening prompt payloads, reusing prepared context, and monitoring unnecessary API calls.
- Built reporting-ready datasets to track AI workflow usage, output quality, exception patterns, adoption signals, latency, retrieval performance, and stakeholder-facing business impact.
- Collaborated with product, data, backend, and business stakeholders to translate ambiguous AI requirements into working prompt behavior, test cases, technical designs, and rollout documentation.
- Documented prompt strategies, API contracts, RAG architecture, evaluation methods, regression tests, failure modes, debugging steps, runbooks, and reusable implementation playbooks.

University of Massachusetts Lowell Sep 2023 – May 2025
Graduate Research Assistant Lowell, MA

- Built AI and ML experimentation workflows using Python, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, OpenCV, SQL, Bash, Jupyter Notebook, and Linux-based environments.
- Designed repeatable data preprocessing pipelines for radar imagery and electrochemical sensor datasets, improving training data reliability by 36% through cleaning, normalization, and validation.
- Developed model evaluation workflows for dataset ingestion, preprocessing, CNN training, metric comparison, error analysis, reporting, and reproducibility documentation.
- Trained and validated convolutional neural network models for contamination detection, achieving RMSE scores of 0.133 for saltwater samples and 0.11 for chalk-infused water samples.

- Applied ML evaluation practices including RMSE tracking, regression-style error analysis, classification-style validation, data-quality checks, and model behavior comparison.
- Used Linux console workflows for environment setup, shell scripting, package configuration, dataset organization, experiment execution, log inspection, and debugging.
- Created technical documentation, experiment designs, configuration notes, dataset assumptions, reproducibility steps, evaluation summaries, and review-ready research reports.

Meesho Jul 2022 – Aug 2023
ML Software Engineer Remote, India

- Built ML-backed data workflows using Python, SQL, Pandas, PySpark, Apache Spark, Scikit-learn, XGBoost, Docker, MongoDB, and Linux-based environments.
- Developed production-style feature engineering pipelines from shopper histories, product metadata, seller records, transaction streams, and user-event data to support ranking and recommendation models.
- Created SQL queries and transformation logic to integrate transaction streams, seller activity, user behavior, product metadata, and model-ready datasets.
- Tuned XGBoost ranking workflows on Spark-based datasets, reducing candidate-selection processing time by 29% for recommendation features.
- Segmented behavioral purchasing signals using Scikit-learn models, increasing recommendation engagement by 18% across active browsing cohorts.
- Built validation scripts for missing values, feature freshness, schema consistency, inference-data quality, ranking inputs, and recommendation-output checks.
- Created analytics datasets to track click-through rate, recommendation performance, campaign performance, model behavior, seller exposure, and user engagement.
- Benchmarked recommendation effectiveness against historical purchasing behavior, improving click-through performance by 14% while maintaining balanced product visibility.
- Collaborated with product, data, and engineering teams in Agile cycles to clarify requirements, troubleshoot pipeline failures, communicate model behavior, and improve production ML reliability.

Projects

AI Character Behavior and Canon RAG Prototype 2025
Python, LangChain, OpenAI APIs, Gemini API, Vector Search, Prompt Evaluation, Guardrails Concepts

- Built a character-focused AI prototype for testing personality consistency, canon grounding, tone rules, safe response boundaries, and multi-turn conversational behavior.
- Designed system prompts that translated character attributes such as personality, voice, tone, boundaries, emotional style, and allowed knowledge scope into testable AI behavior rules.
- Implemented RAG-style retrieval using embeddings, chunked canon documents, vector search, relevance checks, and grounded response generation to reduce unsupported character outputs.
- Created evaluation checklists for character fidelity, canon accuracy, tone consistency, refusal behavior, hallucination risk, safety, and response usefulness across conversation types.
- Designed regression tests to compare prompt versions, model outputs, retrieval quality, and behavior drift after system prompt or provider changes.
- Built adversarial test prompts to challenge boundary handling, off-character responses, unsafe requests, prompt injection attempts, and inconsistent character voice.
- Documented prompt versions, behavior definitions, retrieval assumptions, test cases, evaluation rubrics, failure patterns, and improvement notes for creative and technical review.

Anvikshiki AI 2026
Next.js, React, TypeScript, PostgreSQL, Prisma, Zod, OpenAI API, Gemini API

GitHub: <https://github.com/TarunAngalla/AnvikshikiAI.git>

- Developed an AI-powered coaching SaaS application enabling goal tracking, habit management, reflections, personalized recommendations, structured outputs, and recurring workflow automation.
- Engineered conversational backend workflows for recommendation generation, reflection analysis, habit review, structured JSON validation, user-state tracking, and recurring progress summaries.
- Designed prompt contracts and behavior rules to generate consistent, helpful, context-aware coaching responses across repeated user interactions.
- Architected modular backend services using Prisma, PostgreSQL, and Zod validation, reducing data validation failures by 41% while supporting scalable user profiles and structured memory.
- Implemented AI provider abstraction supporting OpenAI and Gemini models, accelerating model integration cycles by 45% through standardized prompt and response contracts.
- Built dashboard-ready logic for user activity, recommendation quality, workflow progress, goal completion, and AI-generated insight tracking.
- Documented workflow assumptions, schema contracts, provider configuration, prompt behavior, validation rules, deployment notes, and future automation expansion plans.

Career Munzil AI Platform 2025
Python, FastAPI, PostgreSQL, OpenAI APIs, Vector Search, SQL, React

Link: <https://careermanzil.com/>

- Built an AI-enabled SaaS platform supporting resume analysis, skill extraction, candidate recommendations, semantic search, workflow tracking, and stakeholder-facing reporting.
- Designed RAG workflows to compare candidate profiles, extracted skills, role requirements, semantic evidence, and recommendation outputs using vector search and structured prompts.
- Developed FastAPI services for AI inference, candidate matching, skill extraction, structured JSON generation, SQL-backed data

- access, and workflow state management.
- Built validation rules, review checkpoints, exception flags, and output-quality checks to improve trust in LLM-generated summaries and recommendations.
- Created SQL-connected workflows to store extracted skills, scoring outputs, reviewer actions, recommendation records, workflow events, and reporting metrics.
- Documented prompt patterns, API contracts, schema definitions, RAG assumptions, SQL dependencies, test cases, deployment notes, and support handoffs.

Water Quality Monitoring System

2024

Python, TensorFlow, Keras, CNN, NumPy, Pandas, OpenCV, Radar Imaging

GitHub: <https://github.com/TarunAngalla/WaterQualityMonitoring>

- Built a multimodal AI system integrating radar imagery and electrochemical sensor data to identify contamination patterns from structured and image-based inputs.
- Trained convolutional neural networks using TensorFlow and Keras, achieving RMSE scores of 0.133 for saltwater samples and 0.11 for chalk-infused water.
- Designed preprocessing pipelines using Pandas and OpenCV to normalize radar and potentiostat datasets, improving training data reliability by 36%.
- Applied model-evaluation practices including RMSE tracking, error inspection, data-quality review, preprocessing validation, and repeatable experiment documentation.

Badri Digital Strategies Website Platform

2025

React, TypeScript, Node.js, REST APIs, PostgreSQL, Vercel

Link: <https://bds-delta.vercel.app/>

- Built a full-stack business website platform with responsive React interfaces, reusable components, service pages, contact workflows, and production-ready routing.
- Developed backend-ready API workflows for contact submission, lead capture, structured payload validation, service inquiry handling, and future CRM-style integration.
- Designed PostgreSQL-ready data models for leads, service requests, user messages, campaign metadata, and reporting workflows.
- Deployed the platform through Vercel with Git-based version control, environment configuration, build validation, and production-ready frontend optimization.

Education

University of Massachusetts Lowell

Sep 2023 – May 2025

M.S. in Computer Science

Lowell, MA

CVR College of Engineering

Aug 2019 – May 2023

B.Tech in Computer Science and Engineering

Hyderabad, India

Certifications and Achievements

- AWS Certified Developer – Associate
- Microsoft Azure AI Fundamentals
- HackerRank Certifications in Python, SQL, Problem Solving, and Full Stack Development
- Broadridge India Innovation Hackathon Winner for AI-enabled workflow automation systems